TERNARY CODES WITH SYMMETRY AND HOLOMORPHIC SCFTS OF MINIMAL INDEX

[DRAFT — AUTHORS TO BE ADDED]

Abstract. Gaiotto and Johnson-Freyd asked whether for every k there is a holomorphic $\mathcal{N}=1$ vertex operator superalgebra of central charge $12k$ whose supersymmetric index is $24/\gcd(k,24)$, the smallest value allowed by topological modular forms. They answered this for $k \le 5$ by orbifolding the SVOA of a self-dual ternary code of index 24 by a code automorphism of order $\gcd(k,24)$ with prescribed properties, the codes being found by computer search. We make the partial-quadrupling construction of index-24 codes equivariant: it commutes with code automorphisms that permute its blocks, and the twined index of an order-3 automorphism is carried along as a block-weighted index of a collapsed fixed code. Starting from the Golay code and from one computer-found code of length 36 with a symmetry of order 12, we obtain for every k not divisible by 8 an explicit code and automorphism satisfying their hypotheses. This answers the question for all $k \not\equiv 0 \pmod 8$. Conversely, we show that a self-dual ternary code with an automorphism g satisfying $g^4 = -1$ has index divisible by 48, so the same cyclic-orbifold method cannot reach $k \equiv 0 \pmod 8$.

1. Introduction

A self-dual ternary code $C \subseteq \mathbb{F}_3^n$ determines a holomorphic $\mathcal{N}=1$ SVOA $V_{\Lambda(C)}$ of central charge n [2]. Its supersymmetric (Ramond–Ramond) index is

$$\mathrm{Index}(C) = \#\{\text{even maximal codewords}\} - \#\{\text{odd maximal codewords}\},$$

where a codeword is maximal if it has no zero entry and its parity is that of the number of entries equal to 1. The index is defined up to sign (negating a coordinate negates it), and is 0 unless $12 \mid n$. Put $\varphi(0)=0$, $\varphi(1)=-1$, $\varphi(2)=1$ and $f(x)=\prod_i \varphi(x_i)$, so that $\mathrm{Index}(C)=\sum_{c\in C} f(c)$; for any subset $X \subseteq \mathbb{F}_3^n$ write $F(X)=\sum_{x\in X} f(x)$.

By the Stolz–Teichner programme and the computation of the image of $\pi_{24k}\mathrm{TMF} \to \mathrm{MF}_{12k}$ [3], the index of a holomorphic $\mathcal{N}=1$ SCFT of central charge $12k$ should be divisible by $24/\gcd(k,24)$. Gaiotto and Johnson-Freyd ask for the converse:

Main Question ([2])**.** *For each k, does there exist a holomorphic $\mathcal{N}=1$ VOA with central charge $12k$ and supersymmetric index $24/\gcd(k,24)$?*

Their tool is the following theorem. The automorphism group $\mathrm{Aut}(C)$ is taken inside the group $2^n{:}S_n$ of signed coordinate permutations; it always contains -1.

Theorem 1.1 ([2, Thm. 5.6])**.** *Let C be self-dual of length $12k$ with $\mathrm{Index}(C)=24$, put $m=\gcd(24,k)$, and let $g \in \mathrm{Aut}(C)$ have order m. Suppose that (1) if m is even then $g^{m/2}=-1$; and (2) if $3 \mid m$ then $g^{m/3}$ fixes as many even as odd maximal codewords and permutes the coordinates in 3-cycles only. Then g lifts to a non-anomalous automorphism of $V_{\Lambda(C)}$ and the orbifold $V_{\Lambda(C)}/\!\!/\langle g\rangle$ has index $24/m$.*

Negating one coordinate of C and conjugating g by that sign change preserves the hypotheses on g (the parity of every maximal codeword flips at once) and negates the index, so $\mathrm{Index}(C) =$

Date: Draft, 26 September 2026.

± 24 suffices. With codes found by computer search, this answers the Main Question for $k \le 5$ [2, Ex. 5.7]. In [1] we constructed self-dual codes of index 24 in every length $12k$ by a *partial quadrupling* Q_σ (recalled in §2). Here we show how to build the required automorphism into that construction.

Theorem A. *For every $k \ge 1$ with $8 \nmid k$ there is an explicit self-dual ternary code C of length $12k$ with $\mathrm{Index}(C) = \pm 24$ and an automorphism g satisfying the hypotheses of Theorem 1.1. Hence the Main Question has a positive answer for all $k \not\equiv 0 \pmod 8$.*

The residue classes $\gcd(k,24) \in \{1,2\}$ need nothing new (§4.1); $\gcd(k,24) = 4$ uses an automorphism J of the Golay code with $J^2 = -1$ (§4.2); $\gcd(k,24) \in \{3,6,12\}$ uses one code of length 36 with an automorphism of order 12, found by the method of [2] (§4.3). The remaining classes $k \equiv 0, 8, 16 \pmod{24}$ are out of reach of Theorem 1.1:

Theorem B. *Let C be a self-dual ternary code of length $12k$ with an automorphism g such that $g^4 = -1$. Then $48 \mid \mathrm{Index}(C)$, and $\mathrm{Index}(C) = 0$ if k is odd. In particular, when $8 \mid k$ no code of index 24 admits an automorphism as in Theorem 1.1.*

Indeed for $m \in \{8, 24\}$ condition (1) asks for $g^{m/2} = -1$, and then $h = g^{m/8}$ satisfies $h^4 = -1$. We comment on non-cyclic groups in Remark 5.1.

2. Partial quadrupling

Let $T \subseteq \mathbb{F}_3^4$ be the tetracode and $\lambda(0,1,2,3,4) = (0,-2,0,0,1)$. A *block system* σ on a coordinate set X is a set of disjoint 4-subsets (blocks), each split into two pairs; the other coordinates are singletons. For a code $E \subseteq \mathbb{F}_3^X$ put $R_\sigma(E) = \sum_{e \in E} \prod_{i \text{ singleton}} \varphi(e_i) \prod_{\beta \in \sigma} \lambda(\mathrm{wt}(e|_\beta))$. For E self-dual of length m and $|\sigma| = b$, the code $Q_\sigma(E) \subseteq \mathbb{F}_3^{m+12b}$ is defined in [1, Def. 6]: a block $\beta = \{p,q\} \sqcup \{r,s\}$ receives new coordinates β^* carrying a copy of T, each pair (i, i^*) is glued by $h(x,y) = (x+y, x-y)$, the *step-one blocks* $\{p, p^*, q, q^*\}$ and $\{r, r^*, s, s^*\}$ receive new coordinates ** carrying copies of T, and each pair (y, y^{**}) is glued by h again. $Q_\sigma(E)$ is self-dual, and for every set $\mathcal{B}$ of step-one blocks, each split into two pairs $\{y_1, y_2\} \sqcup \{y_3, y_4\}$ and refined into $\{y_1, y_1^{**}, y_2, y_2^{**}\}$ and $\{y_3, y_3^{**}, y_4, y_4^{**}\}$, the resulting block system σ'' satisfies

$$\mathrm{Index}(Q_\sigma(E)) = R_\sigma(E), \qquad R_{\sigma''}(Q_\sigma(E)) = R_\sigma(E) \tag{1}$$

[1, Lemma 8]. The proof of (1) sums over the tetracode variables block by block, so both identities hold with Index replaced by F for *any* linear code E, self-dual or not.

3. Equivariance

Lemma 3.1. *Let $G \subseteq \mathrm{Aut}(E)$ be a finite group that permutes the blocks of σ and maps split pairs to split pairs. Extend each $g \in G$ to the new coordinates by letting i^* follow i with the same sign: if $g(e_i) = \pm e_j$ then $g(e_{i^*}) = \pm e_{j^*}$, and likewise for **. Choose the tetracodes G-equivariantly: on one block of each G-orbit a copy of T invariant under the stabiliser, and its transports elsewhere. Then $G \subseteq \mathrm{Aut}(Q_\sigma(E))$; the extension preserves orders, relations $g^j = -1$, and cycle lengths; and if $\mathcal{B}$ and its splits are G-stable, then σ'' is G-stable.*

Proof. On a glued pair (i, i^*) the extended g acts by a common sign, which commutes with h. So the two gluing maps commute with G, and $Q_\sigma(E)$ is the image of the G-invariant code $E \oplus T^b \oplus T^{2b}$. The remaining statements follow from the definition of the extension. □

A stabiliser-invariant copy of T exists in the cases we need. If J swaps $p \leftrightarrow q$ and $r \leftrightarrow s$ with $J^2 = -1$, say $J(e_0) = e_1$, $J(e_1) = -e_0$, $J(e_2) = e_3$, $J(e_3) = -e_2$, then the tetracode spanned by $(1,1,1,0)$ and $(0,1,2,2)$ is J-invariant. An element of order 3 without fixed coordinates

stabilises no block, since $3 \nmid 4$. In particular, in the presence of such an element all block orbits have size divisible by 3, so $b \equiv 0 \pmod 3$ and $k \bmod 3$ is unchanged along the construction.

Lemma 3.2. *Let $h \in G$ have order 3 and no fixed coordinate, and let $T_h(C) = \sum_{x \in C,\ hx=x} f(x)$. Collapsing h-fixed vectors to one coordinate per h-orbit, let D be the collapsed fixed code of E and $\bar\sigma$ the collapsed block system (one block per h-orbit of blocks). Then*

$$T_h(Q_\sigma(E)) = R_{\bar\sigma}(D), \qquad T_h(Q_{\sigma''}(Q_\sigma(E))) = R_{\bar\sigma}(D) \quad \text{for } G\text{-stable } \sigma''.$$

Proof. After a diagonal sign change, which multiplies Index and T_h by the same sign, h permutes coordinates. An h-fixed vector is constant on h-orbits, and $\varphi^3 = \varphi$, so collapsing preserves f. Since $Q_\sigma(E)$ is the G-equivariant image of $E \oplus T^b \oplus T^{2b}$, its fixed vectors are the images of fixed vectors; the fixed vectors of an h-transported triple of tetracodes form one tetracode after collapsing, and the gluing maps commute with collapsing. Hence the collapsed fixed code of $Q_\sigma(E)$ is $Q_{\bar\sigma}(D)$, and (1) for the (possibly non-self-dual) code D gives both identities. □

4. Proof of Theorem A

4.1. $\gcd(k, 24) \in \{1, 2\}$**.** Take any code of length $12k$ and index ± 24 from [1]. For $m = 1$ take $g = 1$; for $m = 2$ take $g = -1$, which fixes no maximal codeword.

4.2. $\gcd(k, 24) = 4$**.** Label the Golay code G_{12} as in [1]. The signed permutation

$$J = (0\ 1)(2\ 3)(4\ 5)(6\ 7)(8\ 9)(10\ 11), \qquad \text{signs } (-,+,-,+,+,-,-,+,+,-,-,+),$$

meaning $J(e_i) = s_i e_{J(i)}$, lies in $\mathrm{Aut}(G_{12})$ and satisfies $J^2 = -1$ (it is one of 792 such elements of $\mathrm{Aut}(G_{12}) \cong 2.M_{12}$). The block $\beta = \{0, 1\} \sqcup \{2, 3\}$ is J-stable with J-stable pairs, and $R_\beta(G_{12}) = 24$. Apply Lemma 3.1 with $G = \langle J \rangle$. The two step-one blocks are J-stable; refine one of them, split into two J-stable pairs $\{p, Jp\} \sqcup \{p^*, (Jp)^*\}$. The result is a system of two J-stable blocks with J-stable pairs, and the construction iterates, adding 24 coordinates per step. By (1) and Lemma 3.1 we get, for every even k, a self-dual code of length $12k$ with Index $= \pm 24$ and $J \in \mathrm{Aut}$ with $J^2 = -1$. For $\gcd(k, 24) = 4$, J has order $4 = m$ and $J^{m/2} = -1$, so Theorem 1.1 applies.

4.3. $\gcd(k, 24) \in \{3, 6, 12\}$**.** On 36 coordinates let g be the product of the six signed 6-cycles $e_{6j+i} \mapsto e_{6j+i+1}$ $(0 \le i < 5)$, $e_{6j+5} \mapsto -e_{6j}$. Then g has order 12, $g^6 = -1$, $J = g^3$ pairs the coordinates within each 6-cycle, and $h = g^4$ has order 3 with cycle shape 3^{12}. Following the search method of [2] (random g-invariant self-dual codes, keeping those with index ± 24 and $T_h = 0$), we found the code E_{36} of Appendix A, with $\mathrm{Index}(E_{36}) = 24$ and $T_h(E_{36}) = 0$; its collapsed fixed code D has length 12 and dimension 6.

A g-stable block is a union of two J-pairs from different 6-cycles; its h-orbit is a g-stable system of three blocks. For the systems

$$\sigma_3 = \{(24, 27|30, 33), (28, 25|34, 31), (26, 29|32, 35)\}$$

(a block $(a, b|c, d)$ is split as $\{a, b\} \sqcup \{c, d\}$) and

$$\sigma_6 = \sigma_3' \cup \{(12, 15|25, 28), (16, 13|29, 26), (14, 17|27, 24)\},$$
$$\sigma_3' = \{(0, 3|7, 10), (4, 1|11, 8), (2, 5|9, 6)\},$$

direct enumeration gives $R_{\sigma_3}(E_{36}) = R_{\sigma_6}(E_{36}) = -24$ and $R_{\bar\sigma_3}(D) = R_{\bar\sigma_6}(D) = 0$. Apply Lemma 3.1 with $G = \langle g \rangle$, refining at each step one h-orbit of J-stable step-one blocks with J-stable splits; this yields a g-stable system of six blocks and adds 72 coordinates. Starting from σ_3 we reach every $k \equiv 0 \pmod 6$, from σ_6 every odd multiple of 3 other than 3, and $k = 3$ is E_{36} itself. By (1) and Lemmas 3.1 and 3.2, every code in these chains has Index $= \pm 24$,

contains g with $g^6 = -1$ and all cycles of length 6, and has $T_h = 0$. Theorem 1.1 then applies with h for $m = 3$, with g^2 for $m = 6$ ($(g^2)^3 = -1$, $(g^2)^2 = h$), and with g for $m = 12$. □

5. Proof of Theorem B

The condition $g^4 = -1$ forces every coordinate cycle of g to have length 4 and an odd number of sign changes. For a maximal x we get $f(gx) = (-1)^{\#\text{cycles}} f(x) = (-1)^{3k} f(x)$, so for odd k the index vanishes. Let k be even. Then $f \circ g = f$, and $\langle g \rangle$ acts freely on nonzero vectors, since every nontrivial subgroup of $\langle g \rangle \cong \mathbb{Z}/8$ contains $g^4 = -1$. Hence $F(X) \in 8\mathbb{Z}$ for every g-invariant $X \subseteq \mathbb{F}_3^{12k} \setminus \{0\}$.

Structure. Over $\mathbb{F}_3$, $x^4 + 1 = (x^2 + x + 2)(x^2 + 2x + 2)$. On each 4-cycle the space $\mathbb{F}_3^4$ splits as $V_+ \oplus V_-$, the kernels of the two factors evaluated at g. Each is Lagrangian, and they are dual to each other. Hence g-invariant self-dual codes are exactly the codes $C = C_+ \oplus C_-$ with $C_+ \subseteq V_+^{3k} \cong \mathbb{F}_9^{3k}$ an arbitrary $\mathbb{F}_9$-subspace and C_- its annihilator in V_-^{3k}.

A four-code identity. J makes C an $\mathbb{F}_9$-space (J = multiplication by $\sqrt{-1}$), and the ternary form is the real part of a Hermitian form. Let $D \subseteq C$ be J-invariant of codimension 2. Then $D^\perp/D$ is a Hermitian hyperbolic plane over $\mathbb{F}_9$. Its 32 nonzero isotropic vectors form exactly four isotropic $\mathbb{F}_9$-lines, so exactly four J-invariant self-dual codes $C_1, \ldots, C_4$ contain D. Every maximal vector is isotropic, and Poisson summation with $\hat{f} = 3^{n/2} f$ gives $F(D^\perp) = 9F(D)$. Hence

$$\text{Index}(C_1) + \text{Index}(C_2) + \text{Index}(C_3) + \text{Index}(C_4) = 12\, F(D). \tag{2}$$

Induction. Let $D_+ \subset C_+$ be an $\mathbb{F}_9$-hyperplane and $D = D_+ \oplus C_-$. Then $D^\perp/D = L_+ \oplus L_-$ is the sum of the two g-eigenlines, so $C_1 = C$ and $C_2 = D_+ \oplus D_+^\circ$ are the g-invariant codes through D (D_+° the annihilator of D_+ in V_-^{3k}), while g, which acts on $L_\pm$ by the two square roots $\pm\lambda \in \mathbb{F}_9$ of $J = \sqrt{-1}$, maps the isotropic line $\mathbb{F}_9(u + u')$ ($u \in L_+$, $u' \in L_-$) to the isotropic line $\mathbb{F}_9(u - u')$ and so swaps C_3 and C_4. As g preserves f, $\text{Index}(C_3) = \text{Index}(C_4) \in 24\mathbb{Z}$ [2, 1]; and $F(D) \in 8\mathbb{Z}$. Now (2) gives $\text{Index}(C) + \text{Index}(C_2) \equiv 0 \pmod{48}$. Removing one dimension of C_+ at a time reaches $C_+ = 0$, i.e. $C = V_-^{3k}$, a direct sum of tetracodes with index 0. Hence $48 \mid \text{Index}(C)$. □

Remark 5.1. Theorem B concerns cyclic groups. The quaternion group $Q_8 = \langle J, K \rangle$ with $J^2 = K^2 = -1$, $JK = -KJ$ has no element of order 8, and random Q_8-invariant self-dual codes do reach index ± 24 (e.g. 27 of 100 at length 24). Lemma 3.1 applies to Q_8 as well: from such a code of length 24 with a Q_8-stable two-block system of value $R = 24$ we obtain Q_8-invariant codes of index ± 24 at every $k \equiv 0 \pmod 4$ (checked explicitly up to length 96). In Q_8 every commuting pair generates a cyclic group, and every nontrivial one contains -1, which fixes no maximal codeword. So the twisted-sector argument of [2] gives index $24/8 = 3$ for $V_{\Lambda(C)} /\!\!/ Q_8$, *provided the Q_8-action is non-anomalous.* Its restrictions to the cyclic subgroups are non-anomalous when $4 \mid k$, but these restrictions need not detect the full anomaly. Deciding this would settle $k \equiv 8, 16 \pmod{24}$; similarly $Q_8 \times \mathbb{Z}/3$ for $k \equiv 0 \pmod{24}$.

Remark 5.2 (Verification)**.** All statements were checked by computer. The chains of §4.2 and §4.3 were built explicitly up to lengths 144 and 108: invariance, $g^6 = -1$ (resp. $J^2 = -1$), cycle shapes and self-duality at every step; $|\,\text{Index}\,| = 24$ at lengths 48 and 72; and $T_h = 0$ by enumerating the fixed codes at lengths 72, 108 and 144. Identity (2) was checked sign-exactly on random configurations at length 24, the induction step at lengths 24 and 48, and $48 \mid \text{Index}$ on random codes with $g^4 = -1$ at lengths 24 to 72. Automorphism groups were computed with nauty/Traces [4]: the index-24 codes of [1] of lengths 36 and 72 have automorphism groups of 2-power order, so they do not satisfy condition (2); the one of length 48 does satisfy Theorem 1.1 with an element J, $J^2 = -1$. Code and data are available at [repository].

Appendix A. The seed code E_{36}

Generator matrix of E_{36} (coordinates $0, \dots, 35$, entries in $\{0, 1, 2\}$):

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100000000000000000101201221101112102
010000000000000000210120222110111210
001000000000000000021012022211011121
000100000000000000102101202221201112
000010000000000000210210220222120111
000001000000000000210211220222120211
000000100000000000101222200220000211
000000010000000000110122020022200021
000000001000000000111012102002220002
000000000100000000111101110200122000
000000000010000000211110011020012200
000000000001000000021111001102001220
000000000000100000112012221112122212
000000000000010000111201122111112221
000000000000001000211120212211211222
000000000000000100021112221221121122
000000000000000010102111222122112112
000000000000000001210211122212111211
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